

Project: High-Availability Web Infrastructure on Azure

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This document outlines the architecture, implementation steps, and key takeaways from deploying a load-balanced, highly available web environment using Azure Standard Load Balancer and Azure Bastion.

1. Architecture Overview

The solution consists of a multi-tier network architecture designed for security and reliability. Two Windows Server VMs are placed in an Availability Set behind a Standard Load Balancer. Management access is secured via Azure Bastion, eliminating the need for public RDP ports.

Component	Service Used	Purpose
Load Balancer	Azure Standard LB	Traffic distribution and Health Probing.
Compute	Azure VMs (Windows Server)	Hosting IIS Web Services.
Availability	Availability Set	Protection against hardware/update failures.
Secure Access	Azure Bastion (Standard)	Zero-exposure RDP management.
Outbound Traffic	Outbound Rules (SNAT)	Enabling internet access for private VMs.

2. Key Implementation Steps

- **Networking:** Configured a VNet with a dedicated AzureBastionSubnet.
- **Load Balancing:** Established Front-end IP, Backend Pools, and Health Probes (Port 80).
- **Security:** Configured Network Security Groups (NSG) to allow HTTP (80) while keeping RDP (3389) closed to the internet.
- **Provisioning:** Used three different methods to configure VMs:
 1. Server Manager (GUI) for manual feature exploration.
 2. PowerShell (CLI) for rapid automated deployment.
 3. Serial Console for "Out-of-Band" emergency configuration.

3. Lessons Learned & Troubleshooting

Defense in Depth (The Double Firewall)

One major takeaway was that enabling traffic in the Azure NSG is not enough. The Guest OS (Windows) maintains its own internal firewall. For the web service to be reachable, Port 80 had to be opened at both the **Network Layer (NSG)** and the **OS Layer (Windows Firewall)**.

Outbound Connectivity

Unlike Basic Load Balancers, Standard LBs are "Secure by Default." VMs do not have internet access until an **Outbound Rule** is explicitly configured to handle Source Network Address Translation (SNAT).

Management Modernization

With the deprecation of legacy RDP clients, utilizing the **Windows App** and **Azure Bastion** demonstrated a modern, security-first approach to cloud administration.

4. Technical Snippets

Automated IIS Installation and Page Customization:

```
Install-WindowsFeature -name Web-Server -IncludeManagementTools
$vmName = hostname
Set-Content -Path "C:\inetpub\wwwroot\iisstart.htm" -Value "<h1>Lab Success: This is
$vmName</h1>"
New-NetFirewallRule -DisplayName "Allow-HTTP" -Direction Inbound -LocalPort 80
-Protocol TCP -Action Allow
```

5. Final Result

The lab successfully demonstrated a failover scenario. When one VM was deallocated, the Load Balancer Health Probe detected the failure and rerouted traffic to the secondary healthy instance within seconds, maintaining 100% uptime for the application.